

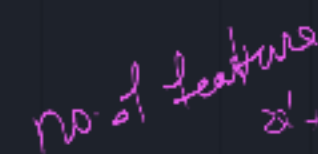
- Simple linear regression
- only one i/p feature
  - multiple i/p features.
  - polynomial (multiple features with higher degree)
- 

## Regularisation

$L_1$  = Lasso

$L_2$  = Ridge

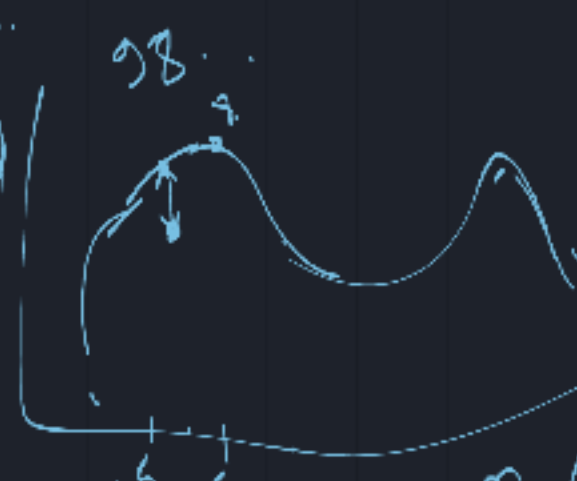
now



Testing ↓

Training  
Testing

100

$$y_{\text{pred}} = \text{lr.predict}(x_{\text{train}})$$


| train |   | red training |
|-------|---|--------------|
| X     | y | $\hat{y}$    |
| 1     | 7 | 7.5          |

testing: 2 8 acc = 80%.

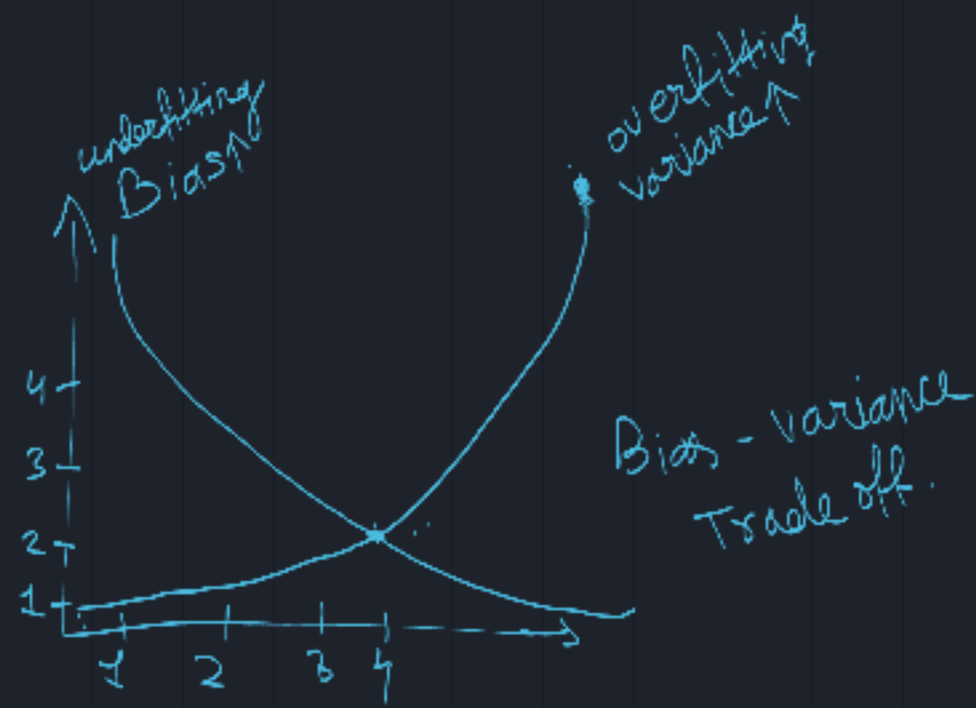
underfitting  $\Rightarrow$  increase no. of feature / increase complexity

no. of feature  $\Rightarrow$  complexity.

(Training acc  $\downarrow$   
Testing acc  $\downarrow$ )

Overfitting  $\Rightarrow$  decrease no. of feature / decrease complexity

(Training acc  $\uparrow$   
Testing acc  $\downarrow$ )



Regularisation algorithm is used for reducing under & overfitting situation  
 → we have two types of Regularisation:-

1] Lasso ( $L_1$ )

2] Ridge ( $L_2$ )

$$L_1 = \frac{\sum (y_i - \hat{y}_i)^2}{n} + \frac{\lambda \sum |\theta|}{1 \quad 2}$$

hyperparameter      coefficient

$$L_2 = \frac{\sum (y_i - \hat{y}_i)^2}{n} + \frac{\lambda \sum \theta^2}{1 \quad 2}$$

learning